Bishop's University
Department of Computer Science
CS503– Data Visualization
Winter 2024 – Project

Objective: Comparative study of Dimensionality Reduction Techniques and their Impact on Regression and Visualization.

Dataset:

1. The dataset is stored in a CSV file named **'diabetes2.csv'**, which has been provided to you.
2. The dataset consists of observations on 442 patients, with the response of interest being a quantitative measure of disease progression one year after baseline. There are ten (10) baseline input variables, **age**, **sex**, **body**-**mass index**, **average blood pressure**, and six blood serum measurements. The last variable '**Y**' is the output.

Tasks:

1. Load the dataset from the CSV file into a DataFrame named **diabetes_df** using the Pandas library.
2. Data Preprocessing:
    a. Preprocess the **diabetes_df** by scaling all the variables to the range [0,1] using **MinMaxScaler**.
    b. Convert the scaled data back to a DataFrame named **diabetes_df_s** for easier visualization.
3. Compute the variance of each input variable.
4. Plot the bar chart showing the variances computed in step 4.
5. Generate a heatmap to visualize the pair-wise correlation between the variables (input and output variables).
6. Rank the input variables in descending order based on their correlation with the output variable. The higher the variance, the more important the input variable is.
7. Using the first two important input variables, generate a scatter to display the data distribution.
8. Apply **Lasso regression** to the entire dataset (using all variables).
    a. Lasso regression involves a regularization parameter, denoted as alpha ∝ in the Scikit-learn ML tool. A higher value of alpha (also known as lambda) leads to more regularization, which in turn shrinks the coefficients towards zero, effectively reducing the complexity of the model and selecting only the most important variables.
    b. Using Mean Squared Error (MSE) to calculate the average squared difference between the predicted and actual values. Lower MSE values indicate better model performance. Scikit-learn provides a function for calculating MSE.
    c. Compute the MSE of Lasso regression for different values of alpha: 0, 1, 10, 100, 500, and 1000.
    d. Plot the curve showing the variation of MSE with respect to alpha.

    e. Display the best MSE and the corresponding alpha value.
    f. Plot the evolution of Lasso coefficients against alpha to observe how they change and how they are Shrunk as alpha varies.

9. Reduce the data dimensionality using PCA (Principal Component Analysis).
    a. Utilize PC1 and PC2 and visualize the data scatter.
    b. Plot the loadings to examine how the variables contribute to PC1 and PC2.
    c. Perform **normal linear regression**, using PC1 only.
    d. Plot the regression line on the scatter.
    e. Perform **normal linear regression**, using PC1 and PC2.
    f. Plot the regression hyper-line on the scatter.
    g. Using bar chart, calculate, and display the MSE for both cases 9.c and 9.d.
10. Reduce the data dimensionality with t-SNE.
    a. Utilize the 1st and 2nd t-SNE dimensions to visualize the data scatter, with different **perplexity** values: 5, 10, 20, and 50.
    b. Perform **normal linear regression**, using only the 1^{st} dimension of t-SNE.
    c. Plot the regression line on the scatter.
    d. Perform **normal linear regression**, using the 1^{st} and 2^{nd} dimensions of t-SNE.
    e. Plot the regression hyper-line on the scatter.
    f. Using bar chart, calculate, and display the MSE for both cases 10.b and 10.d.
11. Reduce the data dimensionality with UMAP.
    a. Utilize the 1st and 2nd UMAP dimensions to visualize the data scatter, with different **n_neighbors** (number of neighbors) values: 5, 10, 20, and 50.
    b. Perform **normal linear regression**, using only the 1^{st} dimension of UMAP.
    c. Plot the regression line on the scatter.
    d. Perform **normal linear regression**, using the 1^{st} and 2^{nd} dimensions of UMAP.
    e. Plot the regression hyper-line on the scatter.
    f. Using bar chart, calculate, and display the MSE for both cases 11.b and 11.d.
    g. Provide a comparative table to compare Linear Regression applied to PCA, t-SNE, and UMAP data, utilizing the first three dimensions for each dimensionality reduction method.
12. **Report**:

    Prepare a report spanning 3 to 4 pages, outlining the results, along with detailed discussion and analysis. A template of the report is given in the next pages.

    Five (5) marks will be given to the quality of the report (layout, organization, coherence, etc.).
13. **Note**:
    - All the code should be provided in a Jupyter notebook. You are free to use either Python or R, although Python is more recommended.
    - For each question, create ONE Jupyter CODE CELL. Include a CELL TEXT just before any CODE CELL to indicate the question number. Failure to comply with this requirement will result in a zero mark. Clarity in your program makes marking easier and significantly impacts the results.
    - Any instances of plagiarism or inappropriate use of AI-based tools to solve the problem will result in a zero mark. Not that we have tools for plagiarism detection.

Title (less than 15 words)

Your name
Your email address
Department of Computer, Bishop's University

Abstract

Provide an abstract of the word to present. It is about a comparative study of different dimensionality reduction methods on regression.

1. **Introduction**:

   The paragraph presents the context of the problem to solve. It is about 10 lines.

   Objectives:

   What is the objectives to achieve. About 3 lines.

2. **Background**:

   About 1 page and half. Describe data visualization, regression, dimensionality reduction methods emphasizing the theory of the method (not in details).

3. **Experiments and evaluation**:
   3.1. Describe the dataset.
   3.2. Using graphs, and tables show the results of each DR method and discuss how the different parameters affect regression results.
   3.3. Using graphs and tables, compare the best results obtained by each method.
   3.4. Discuss the finding.
4. **Conclusion**:
   About 7 lines. Conclude your work.